photoelectric effect even if the frequency is less than the threshold frequency. So the concept of threshold frequency does not even exist in wave theory.

ii) According to wave theory, the intensity of a light wave is proportional to the square of the amplitude of the electric field (E_0^2). The amplitude of this electric field increases with increasing intensity and imparts an increasing acceleration and kinetic energy to an electron.

Now I is replaced with a quantity proportional to E_0^2 in equation (1). This means that K_{max} should not depend at all on the frequency of the classical light wave which again contradicts the experimental results.

iii) If an electron accumulates light energy just enough to overcome the work function, then it is ejected out of the atom with zero kinetic energy. Therefore, from equation (1),

$$0 = IAt - \phi_0$$

$$t = \frac{\phi_0}{IA} = \frac{\phi_0}{I(\pi r^2)}$$

By taking the atomic radius $r = 1.0 \times 10^{-10}$ m and substituting the given values of I and ϕ_0, we can estimate the time interval as

$$t = \frac{2.14 \times 1.6 \times 10^{-19}}{1.60 \times 10^{-6} \times 3.14 \times (1 \times 10^{-10})^2}$$

$$= 0.68 \times 10^7 \, s \approx 79 \text{ days}$$

Thus, wave theory predicts that there is a large time gap between the incidence of light and the ejection of photoelectrons but the experiments show that photo emission is an instantaneous process.

Concept of quantization of energy

Max Planck proposed quantum concept in 1900 in order to explain the thermal radiations emitted by a black body and the shape of its radiation curves.

According to Planck, matter is composed of a large number of oscillating particles (atoms) which vibrate with different frequencies. Each atomic oscillator - which vibrates with its characteristic frequency - emits or absorbs electromagnetic radiation of the same frequency. It also says that

i) If an oscillator vibrates with frequency ν, its energy can have only certain discrete values, given by the equation.

$$E_n = nh\nu \qquad n=1,2,3.... \qquad (8.5)$$

where h is a constant, called Planck's constant.

ii) The oscillators emit or absorb energy in small packets or quanta and the energy of each quantum is $h\nu$.

This implies that the energy of the oscillator is quantized – that is, energy is not continuous as believed in the wave picture. This is called **quantization of energy**.

8.2.7 Particle nature of light: Einstein's explanation

Einstein extended Planck's quantum concept to explain the photoelectric effect in 1905. According to Einstein, the energy in light is not spread out over wavefronts but is concentrated in small packets or energy quanta. Therefore, light (or any other electromagnetic waves) of frequency ν from any source can be considered as a stream of quanta and the energy of each light quantum is given by $E = h\nu$.

He also proposed that a quantum of light has linear momentum and the magnitude of that linear momentum is $p = \frac{h\nu}{c}$. The individual light quantum of definite energy and momentum can be associated with a

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